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DevOps Lab Experiment – 7

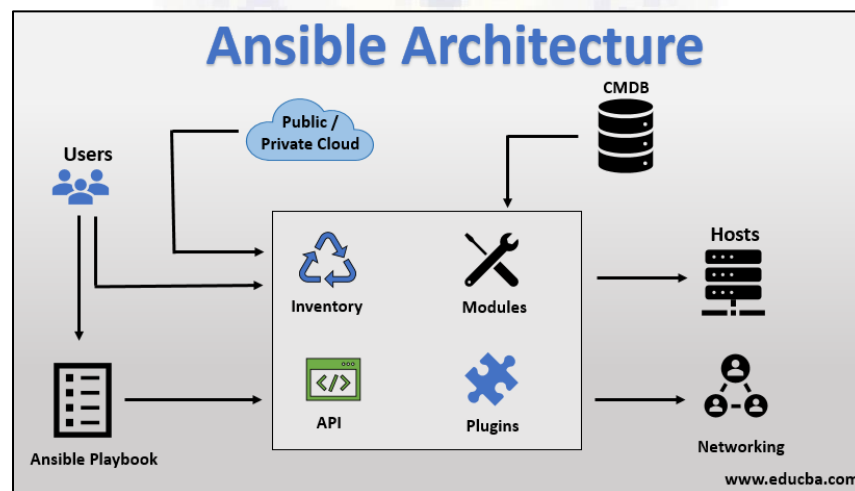
Configuration Management with Ansible: Basics of Ansible: Inventory, Playbooks, and Modules, Automating Server Configurations with Playbooks, Hands-On: Writing and Running a Basic Playbook.

Overview of Ansible

Ansible is an open-source configuration management tool, written in Python for configuration management, application deployment, task automation, and more. Michael DeHaan created Ansible in 2012, and it gained popularity in 2013. Red Hat acquired Ansible in 2015. Nowadays, a large community still contributes to its development.

Ansible Architecture

The **Ansible Automation Engine** is the central component of Ansible architecture. It acts as the **control system** that connects users, infrastructure, and external services.



Ansible Architecture

The Ansible automation architecture consists of several components that work together to automate system configuration and management.

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The **inventory** contains the list of hosts (systems) such as servers, databases, or devices that need to be managed, along with their details like IP addresses. Ansible connects to these systems using protocols such as SSH for Linux/Unix systems and WinRM for Windows systems.

Playbooks are YAML files that define the tasks to be executed on these hosts, while modules are small programs that perform specific actions like installing packages, managing services, or creating files.

Plugins extend Ansible's functionality by adding extra features such as caching or executing tasks on the control machine.

Ansible can also interact with cloud services using **APIs** to create and manage resources, and it can automate network devices using the same framework. The target systems, known as hosts, can be any machines such as Linux or Windows systems.

Additionally, Ansible can integrate with a **CMDB (Configuration Management Database)** to store and retrieve information about system configurations.

Overall, Ansible acts as a central automation engine that connects users, infrastructure, cloud services, and databases to perform tasks efficiently.

1. Interaction with Users (via Playbooks)

- Users write automation instructions in the form of playbooks (YAML files)
- These playbooks define:
 - Target hosts
 - Tasks to execute
 - Modules to use

2. Interaction with Cloud Services

Ansible integrates with cloud platforms such as AWS, Azure, and GCP.

The engine can

- Create virtual machines
- Configure cloud resources
- Deploy applications

3. Interaction with CMDB (Configuration Management Database)

CMDB stores information about:

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- Servers
- IP addresses
- System configurations

Ansible uses CMDB to

- Fetch host details dynamically
- Update infrastructure data
- Maintain consistency

4. What Happens Internally

1. User runs playbook
2. Ansible Engine
 - Reads **inventory**
 - Loads **modules**
 - Uses **plugins/API**
3. Connects to
 - Hosts (via SSH or local)
4. Executes tasks step-by-step
5. Returns output to user

The Ansible Automation Engine acts as a bridge between users and infrastructure. It simplifies complex operations by automating tasks across local systems, cloud environments, and enterprise databases like CMDB.

Advantages of Using Ansible

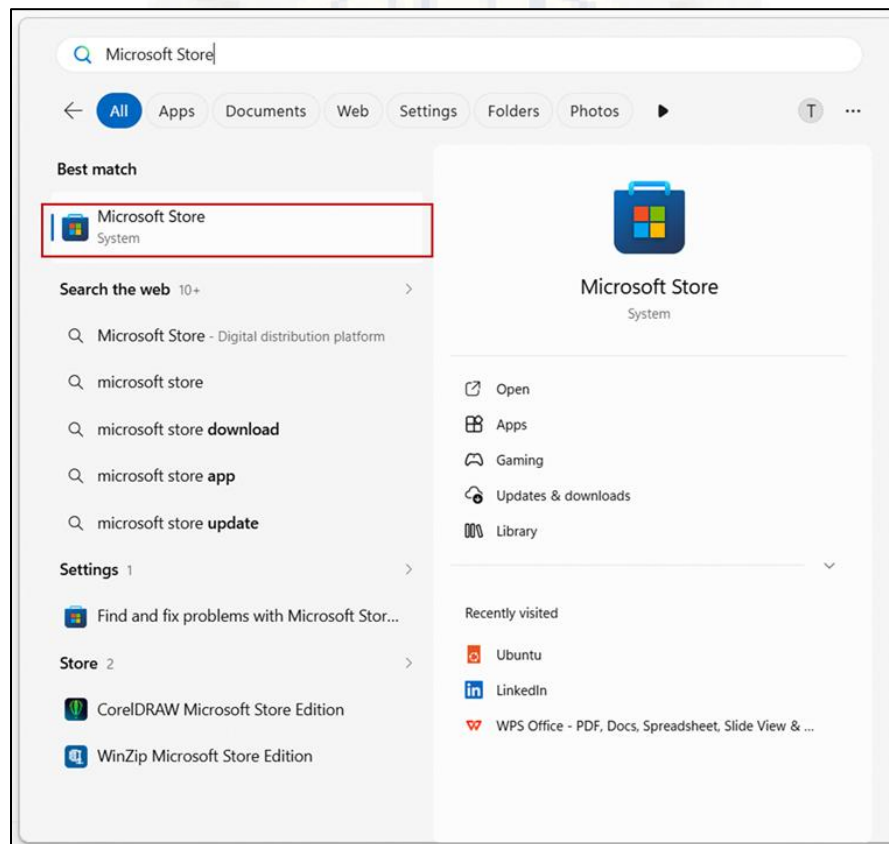
- **Free and Open Source** : Ansible is an open-source tool, so it is freely available and widely supported by the community.
- **Simple to Set Up and Use** : Ansible is easy to install and use. It uses simple YAML-based playbooks, so no advanced coding skills are required.
- **Powerful** : Ansible can handle complex IT workflows, including multi-tier application deployments and large-scale automation.

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- **Flexible :** It can manage and automate the entire application environment, whether it is on premises, in the cloud, or hybrid. It is also highly customizable.
- **Agentless Architecture :** Ansible does not require any agent installation on client systems. It works over SSH, reducing setup complexity and security risks.
- **Efficient Resource Usage :** Since no additional software is installed on managed nodes, system resources are saved, improving performance.

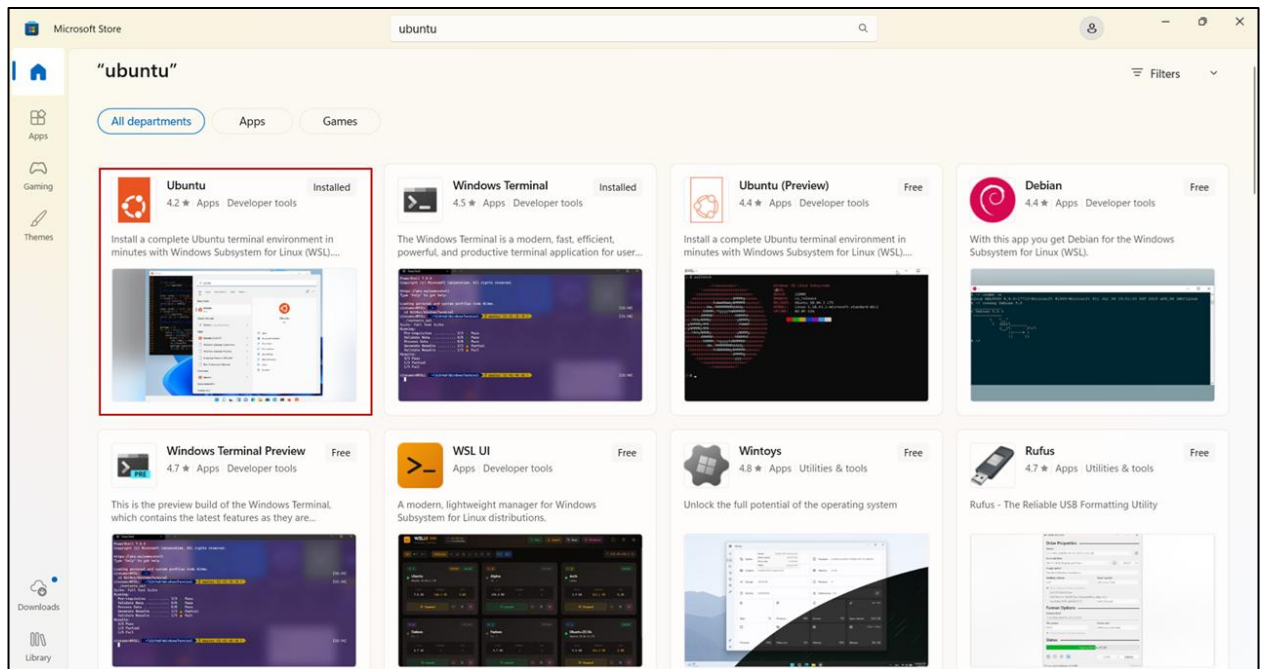
Step 1: Ansible Installation on Windows Using Windows Subsystem for Linux (WSL)

- Click on the Start Menu → Type **Microsoft Store** in the search bar → Select and open the **Microsoft Store** application

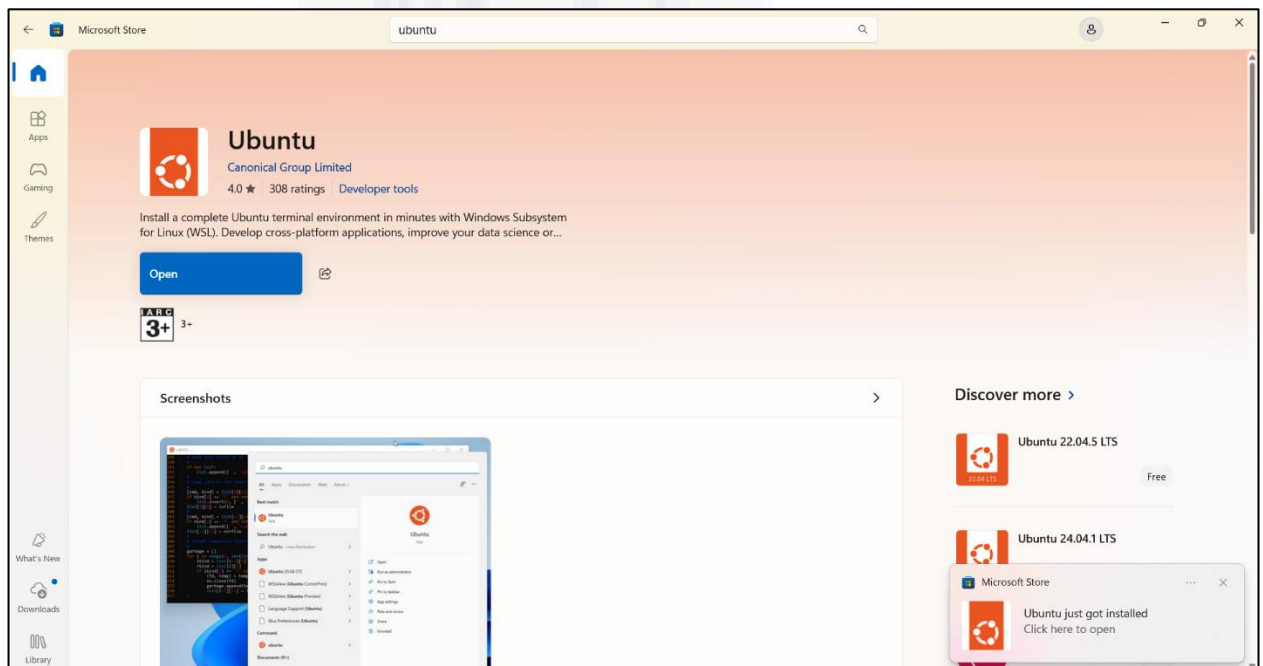


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- In the Microsoft Store search bar, type **Ubuntu** → Install the Latest LTS Version

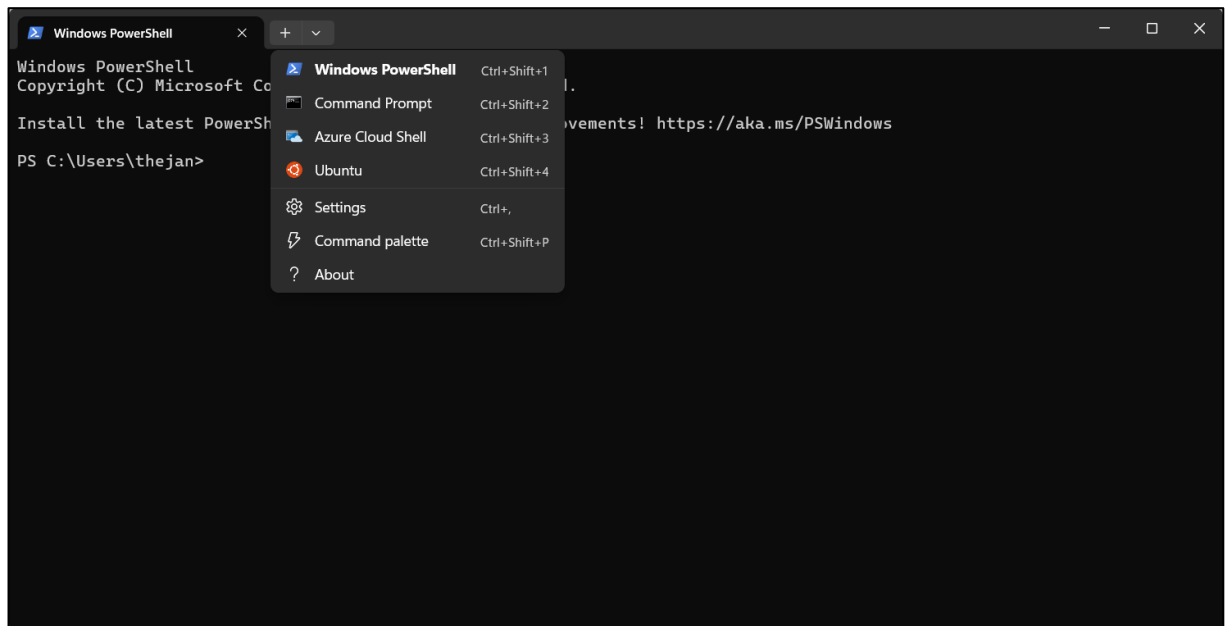


- After installation, click on the **Open** button. Ubuntu will launch and complete the setup process



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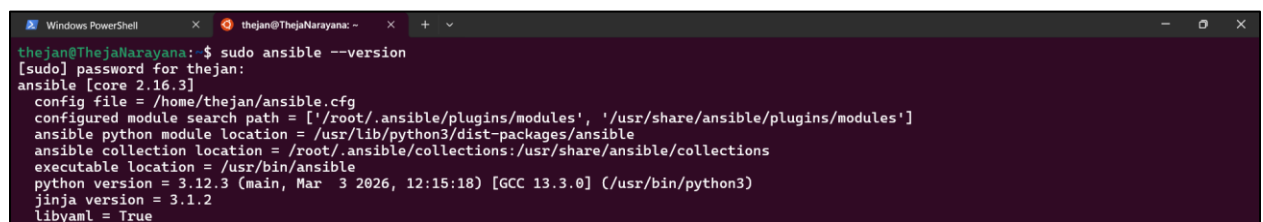
- Open **Windows Terminal** → Click on the **dropdown arrow** (↓) next to the tab and select **Ubuntu** from the list. This opens the Ubuntu environment inside Windows Terminal using **WSL (Windows Subsystem for Linux)**.



- Set a new UNIX username (admin) and password (admin123) when prompted.
- Run the following commands in the terminal to update the package list and upgrade installed packages:

```
sudo apt update  
sudo apt upgrade -y
```

- Run the command **sudo apt install ansible** -y to install Ansible.
- Verify the installation by checking its version using the command : **ansible --version**



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If **config file = None (config file is not found)**, then follow these steps

- Create a configuration file using the command **nano ansible.cfg**
- Save and Exit : Press **Ctrl + O → ENTER → Ctrl + X**
- Verify the configuration by checking the version

Step 2: Creating an Ansible Inventory

- Create an **Ansible inventory file** using the command **nano hosts.ini**
- Add the following content to define the local host

```
[local]
localhost ansible_connection=local
```

- Save and Exit : Press **Ctrl + O → ENTER → Ctrl + X**

Step 3 : Create the Ansible Playbook File

- Create the **Ansible playbook file** using the command **nano setup.yml** and add the following content. **Save (Ctrl + O) → ENTER → Exit (Ctrl + X)**.

```
---
- name: Install and start Nginx
  hosts: local
  become: yes

  tasks:
    - name: Install Nginx
      apt:
        name: nginx
        state: present
        update_cache: yes

    - name: Ensure Nginx is running
      service:
        name: nginx
        state: started
        enabled: yes
```

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- Run the playbook file using the command

ansible-playbook -i hosts.ini setup.yml --ask-become-pass

and enter the password.

```
Windows PowerShell
thejan@ThejaNarayana: ~
thejan@ThejaNarayana: $ ansible-playbook -i hosts.ini setup.yml --ask-become-pass
BECOME password:

PLAY [Install and start Nginx] *****
TASK [Gathering Facts] *****
ok: [localhost]
TASK [Install Nginx] *****
changed: [localhost]
TASK [Ensure Nginx is running] *****
ok: [localhost]
PLAY RECAP *****
localhost : ok=3  changed=1  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
```

Modify hosts.ini and setup.yml as shown below and run the playbook.

```
[webservers]
localhost ansible_connection=local
```

```
---
- name: Install and start Nginx
  hosts: webservers
  become: yes

  tasks:
    - name: Install Nginx
      apt:
        name: nginx
        state: present
        update_cache: yes

    - name: Ensure Nginx is running
      service:
        name: nginx
        state: started
        enabled: yes
```


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```
Windows PowerShell
thejan@ThejaNarayana: ~
thejan@ThejaNarayana: $ ansible-playbook -i hosts.ini setup.yml --ask-become-pass
BECOME password:

PLAY [Install and start Nginx] *****
TASK [Gathering Facts] *****
ok: [localhost]
TASK [Install Nginx] *****
ok: [localhost]
TASK [Ensure Nginx is running] *****
ok: [localhost]
PLAY RECAP *****
localhost : ok=3  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
```

VIVA QUESTIONS

1. What is Ansible?
2. Is Ansible agent-based or agentless?
3. Which programming language is Ansible written in?
4. What protocol does Ansible use for communication?
5. What is an inventory in Ansible?
6. What is a playbook?
7. What are modules in Ansible?
8. What is YAML?
9. Why is YAML used in Ansible?
10. How does Ansible interact with users?
11. What is CMDB?
12. How do you verify Ansible installation?
13. What is the purpose of sudo apt update?
14. What is the purpose of sudo apt upgrade?
15. What is the use of ansible-playbook command?
16. What does --ask-become-pass do?

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17. Why do we use hosts in a playbook?
18. What happens if hosts do not match the inventory?
19. What is the difference between local and webservers?
20. How do you run a playbook?
21. What does config file = None indicate?
22. What are the advantages of Ansible?
23. Why is Ansible considered agentless?
24. How is Ansible different from other configuration tools?
25. What are the limitations of Ansible?
26. What is the difference between inventory and playbook?
27. What is the workflow of Ansible execution?
28. What are real-world uses of Ansible?